Summary in Graph

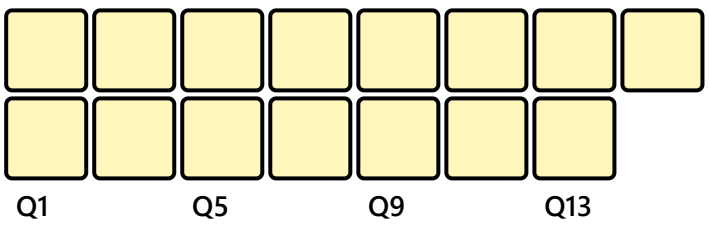
Exam Summary (GO Classes 2026 | Algorithms | DFS BFS Algorithms | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (32) out of 316 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Algorithms

Consider a graph G. Let T be a BFS tree with root r.
Let $d(u, v)$ denote the length of the shortest path between nodes u and v.
If a vertex v is visited before a vertex u in the BFS traversal, which of the following statements is true?

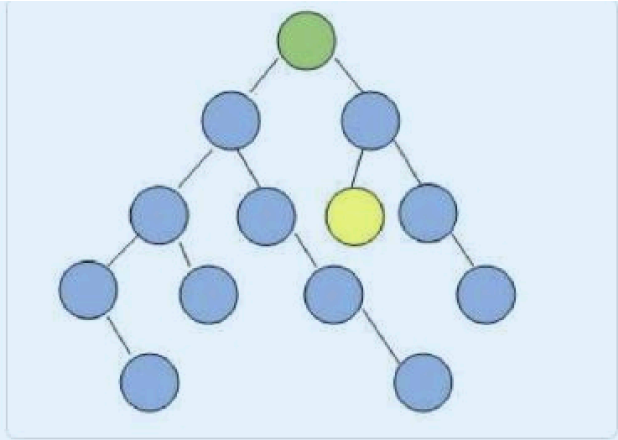
A. $d(r, v) > d(r, u)$
B. $d(r, v) = d(r, u)$

C. $d(r, v) < d(r, u)$
D. $d(r, v) \leq d(r, u)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the tree below. The search starts from the green node at the top. The yellow node is highlighted. Whenever there is a choice among multiple children, both BFS and DFS choose the leftmost child first.



Which algorithm will visit the least number of nodes before reaching the yellow node?

A. BFS
B. DFS
C. Neither BFS nor DFS will ever reach to the Yellow node
D. BFS and DFS visit the same number of nodes before reaching the yellow node

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let G be a connected undirected graph. Suppose we perform both BFS and DFS starting from the same root vertex r. Which of the following is true regarding the depth of the BFS tree and the DFS tree (both rooted at r)?

A. The depth of the BFS tree is always greater than or equal to the depth of the DFS tree.
B. The depth of the BFS tree is always strictly smaller than the depth of the DFS tree.
C. The depth of the DFS tree is always greater than or equal to the depth of the BFS tree.
D. The depths of BFS and DFS trees are always equal.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let G be a simple undirected graph and let T_B be a breadth first search (BFS) tree of G. For every edge (u, v) of G, let $d(u)$ and $d(v)$ denote the depths of u and v in T_B.

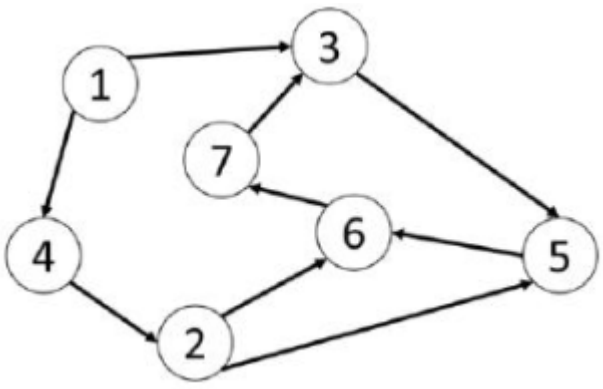
Which of the following must necessarily hold?

A. $|d(u) - d(v)| \leq 1$
B. $|d(u) - d(v)| = 1$
C. $|d(u) - d(v)| \geq 1$
D. $|d(u) - d(v)| = 0$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the given graph shown below:



Start a depth-first search (DFS) from node 1, always visiting the smallestnumbered neighbor first.

Let $d(v)$ denote the discovery time of vertex v, and let $f(v)$ denote its finish time.

Which of the following relations is always true? (select all that apply)

A. $d(3) < d(7) < f(7) < f(3)$
B. $d(4) < d(2) < f(2) < f(4)$
C. $d(5) < d(6) < f(6) < f(5)$
D. $d(7) < d(6) < f(7) < f(6)$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a directed graph with n vertices arranged in a path:

- For each $1 \leq i < n$ there is the forward edge $(i, i+1)$.
- In addition, for every pair $i > j$ there is an edge (i, j).

A depth-first search (DFS) is performed on this graph.
What are the minimum and maximum possible numbers of back edges, depending only on the order in which DFS explores vertices?

A. Minimum $= 0$, Maximum $= n - 1$
B. Minimum $= n - 1$, Maximum $= \frac{n(n-1)}{2}$
C. Minimum $= \frac{n(n-1)}{2}$, Maximum $= n(n-1)/2$
D. Minimum $= n - 1$, Maximum $= n - 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

The discovery and finishing times of vertices in a DFS traversal are:
$(q,1,16),(r,17,20),(s,2,7),(t,8,15),(u,18,19),(v,3,6),(w,4,5),(x,9,12),(y,13,14),(z,10,11)$
If the directed edges given in the options are present in the graph, then which of these would be classified in DFS as a back edge?

A. (y,x)
B. (z,y)
C. (w,s)
D. (u,q)

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

The discovery and finishing times of vertices in a DFS traversal are:
$(q,1,16),(r,17,20),(s,2,7),(t,8,15),(u,18,19),(v,3,6),(w,4,5),(x,9,12),(y,13,14),(z,10,11)$
If all possible directed edges between these vertices are considered, what is the maximum number of cross edges that can exist while remaining consistent with the given discovery and finishing times?

Your Answer: | Correct Answer: 30 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

We perform a DFS on a connected undirected graph G.

- The global time starts at 0 and is incremented by 1 whenever a vertex is discovered or finished.
- For each vertex v, let $d(v)$ denote its discovery time and $f(v)$ its finish time.
- Let T be the DFS tree obtained.

Now consider a vertex u that is a leaf in T.
Which of the following statements are always correct?

A. $f(u) = d(u) + 1$
B. $f(u) < f(\text{parent}(u))$
C. $d(u) > d(\text{parent}(u))$
D. For any two leaves u and v, if $d(u) < d(v)$, then $f(u) < f(v)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let G be a connected undirected graph, and let T be a DFS tree obtained from G.
For each vertex x, let $d(x)$ be its discovery time and $f(x)$ its finish time.
Suppose u is a vertex in G, and v is the first vertex discovered after the DFS has completely finished $\text{exploring}\, u$ (i.e., the next new vertex in traversal order after backtracking from u).

Which of the following statements is always true?

A. v cannot be a leaf of the DFS tree T.
B. u and v belong to different DFS subtrees rooted at some common ancestor.
C. It is possible that $\{u,v\}$ is an edge in G, but such an edge can only be a back edge (never a tree edge).
D. u cannot be the root of the DFS tree.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider an undirected graph $G = (V, E)$ with a BFS tree T rooted at vertex 0 .
The BFS partitions the vertices into layers $L_0, L_1, L_2, \ldots$, where L_k contains all vertices at distance k from the root.

Which of the following statements are true?

A. If two vertices in the same layer are connected by an edge, then the graph contains an odd cycle.
B. If two vertices in consecutive layers are connected by an edge, then the graph contains an even cycle.
C. Skip-level edges (edges between vertices whose layers differ by at least 2) can never exist in BFS layering.
D. If two vertices are at distance k from the root, then they cannot be connected by any edge.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose G is a connected undirected graph. An edge whose removal disconnects the graph is called a bridge. Consider a Depth-First Search (DFS) tree of G. Which of the following statements is true regarding bridges in G ?

A. A bridge can only be a back edge in some DFS tree.
B. A bridge can be either a tree edge or a back edge in DFS.
C. Every bridge must necessarily be a tree edge in every DFS tree of G.
D. A bridge can never appear as a tree edge in a DFS tree.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Identify which of the following statements are correct.

A. When visiting a node v during DFS on an undirected graph, if an adjacent node w is colored **gray**, then this indicates a cycle only if w is not the parent of v.
(Gray means “discovered but not finished”.)

B. For an undirected graph G, it is not possible for the BFS tree to have greater height than G' s DFS tree.

C. The time-complexity for using DFS to count connected components in an undirected graph is $\Theta(V(V+E))$.

D. BFS can be used to tell if an undirected graph G is bipartite by checking if a non-tree edge connects nodes on the same level.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider an n-vertex graph G where both DFS (using recursion) and BFS (using a queue) are started from the same vertex v_1. Match each type of graph in Column A with the correct behavior of DFS recursion depth and BFS maximum queue size in Column B.

Column A (Graph type)	**Column B (Property)**
a. Path graph on n vertices	1. DFS recursion depth ≤ 1
b. Star graph on n vertices (center $= v_1$)	2. DFS recursion depth $= 2$
c. Complete graph on n vertices	3. DFS recursion depth $= n - 1$
	4. BFS queue size ≤ 1
	5. BFS queue size $= n - 1$

The correct matching between the elements in Column A with those of Column B is:

A. $a \to (3,4),\ b \to (1,5),\ c \to (3,5)$
B. $a \to (2,5),\ b \to (3,4),\ c \to (1,5)$
C. $a \to (1,5),\ b \to (2,4),\ c \to (3,5)$
D. $a \to (3,5),\ b \to (2,4),\ c \to (1,5)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

In a directed graph $G = (V, E)$, a mother vertex is a vertex v such that all other vertices in G are reachable from v.
Suppose we are given a vertex v. Which of the following procedures correctly tests whether v is a mother vertex in $O(n + m)$ time, where $n = |V|$ and $m = |E|$?

A. Start DFS from v. If the DFS tree has exactly $n - 1$ tree edges, then v is a mother vertex.
B. Start BFS from v. If BFS reaches all other $n - 1$ vertices, then v is a mother vertex.
C. Start DFS from v. If DFS reaches all other $n - 1$ vertices, then v is a mother vertex.
D. Start DFS from v. If the number of cross edges is zero, then v is a mother vertex.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 01sec | Discuss

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